UNIVERSIDADE DE SÃO PAULO - ICMC

Departamento de Ciências de Computação

Seminário da disciplina SCC5871 – Aprendizado de Máquina

BERT's family model



Grupo N° 6

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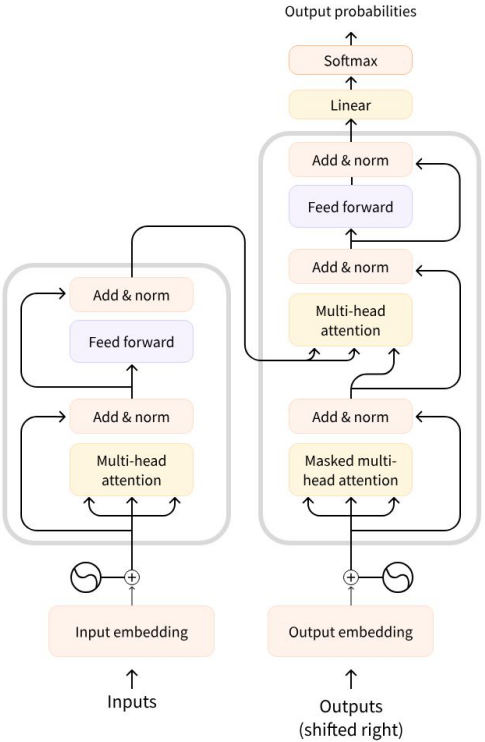
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Introdução

- **Modelos de linguagem antes do BERT**
  - Word2Vec & GloVe
  - LSTMs & GRUs
- **Novidade e Revolução do BERT**
  - MLM
  - NSP
  - Transfer Learning
- **Impacto do BERT**
  - Sistemas de busca
  - Classificação de texto e AS
  - Inspiração de outros modelos

Arquitetura Transformer



Fonte: Vaswani et al. (2017)

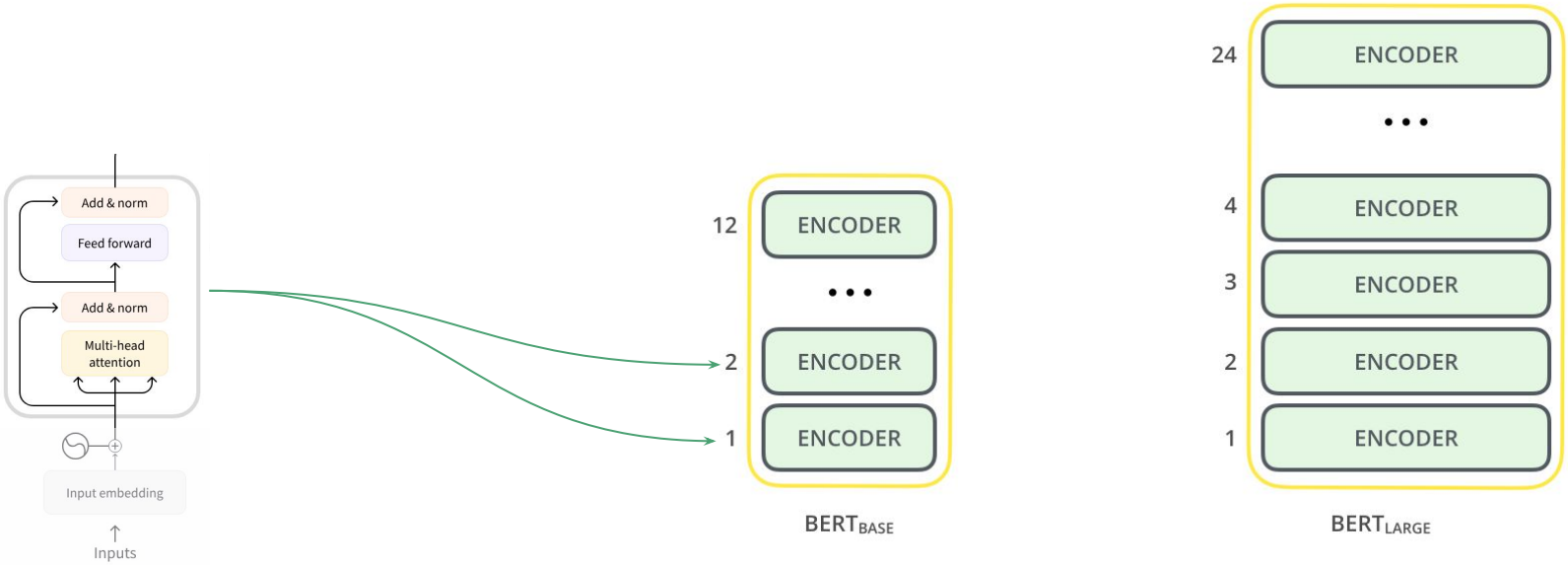
BERT

Bidirectional Encoding Representations from Transformers

Bidirectional Encoding Representations from Transformers - BERT

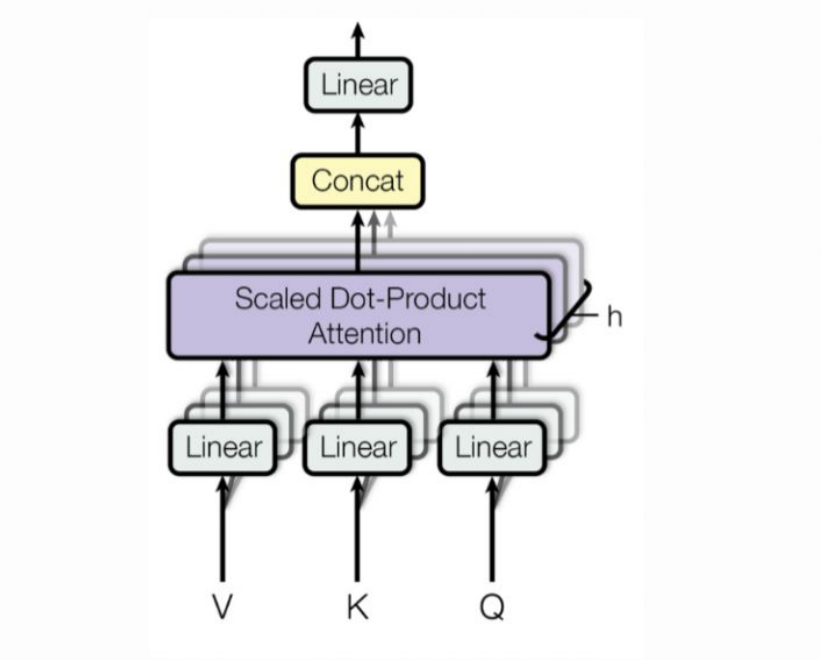
Arquitetura	BERT Base	BERT Large
Camadas ou Transformers blocks	12	24
Self-Attention Heads	12	16
Hidden State Layers	768	1024
Total de parâmetros	110 mil	340 mil

Camadas ou Transformers Blocks



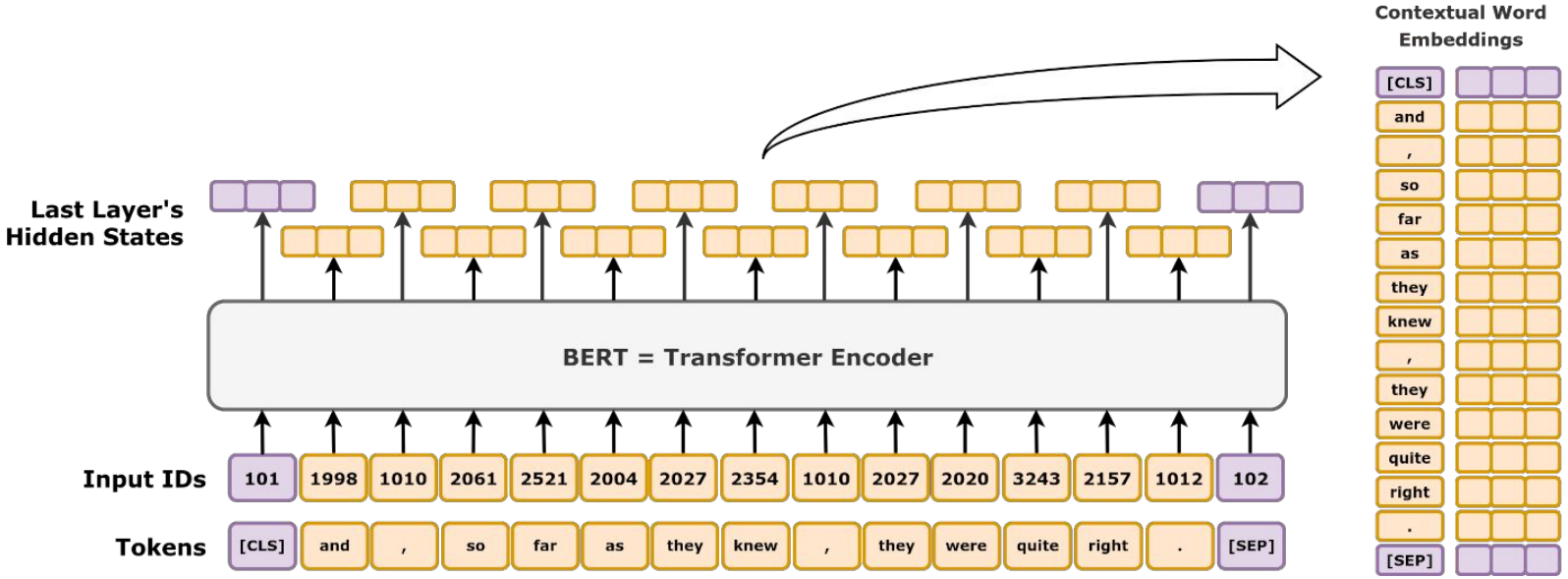
Fonte: Adaptado de Vaswani et al. (2017) e Alammar (2018)

Self-Attention heads



Fonte: Vaswani et al. (2017)

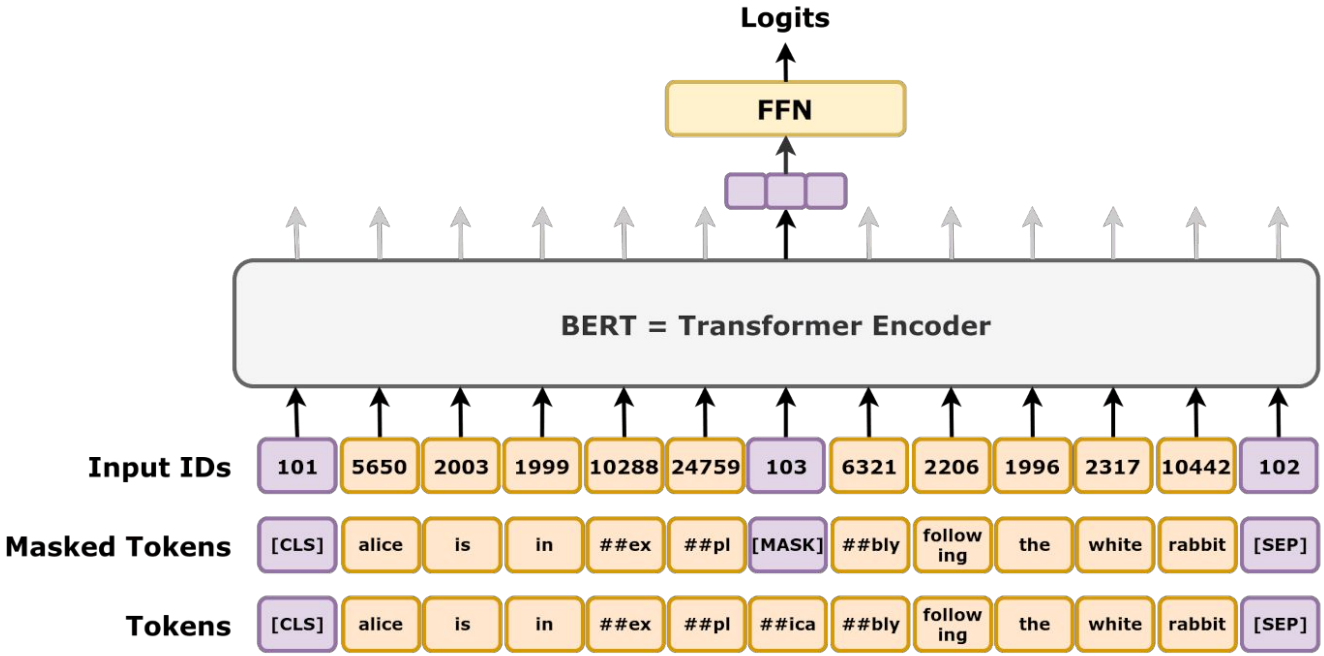
Hidden states



Fonte: Godoy (2019) BERT | dl-visuals

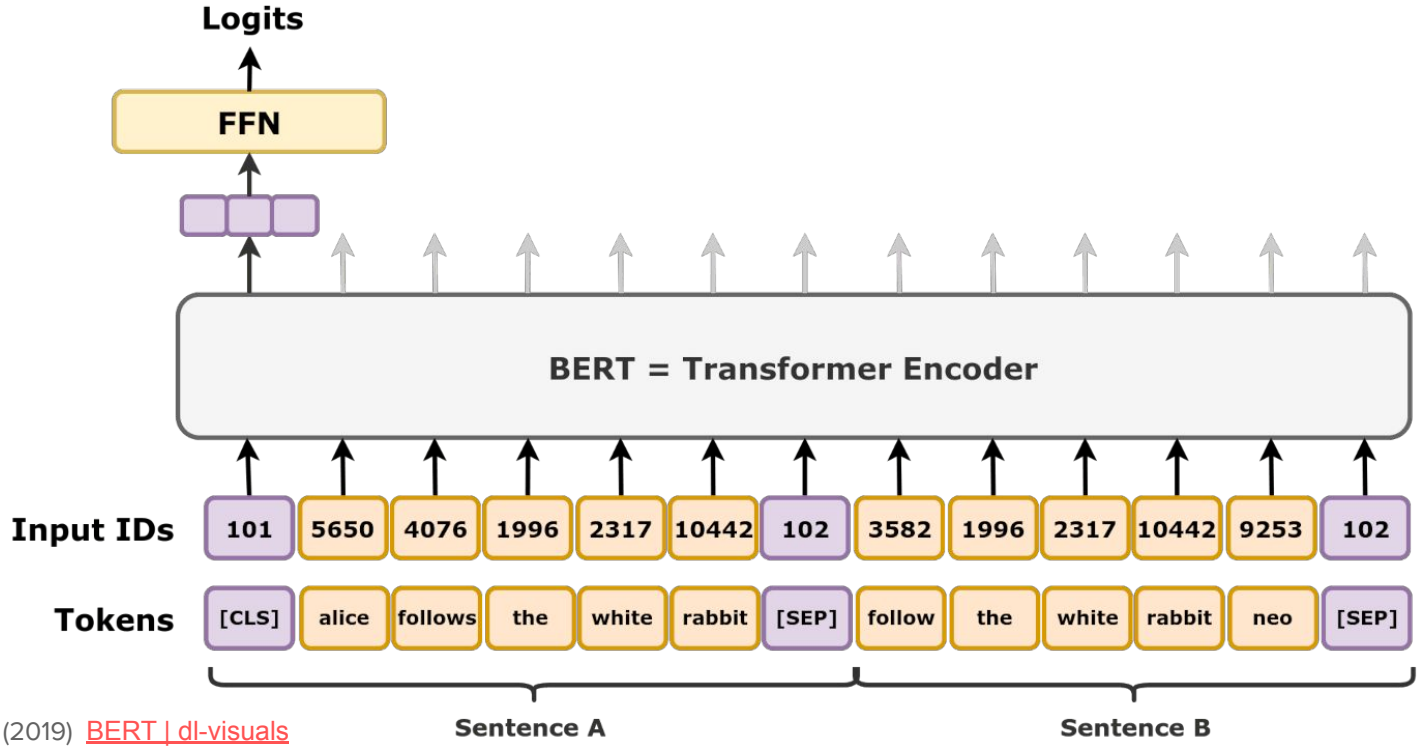
Pré-treinamento

Masked Language Modeling (MLM)



Fonte: Godoy (2019) BERT | dl-visuals

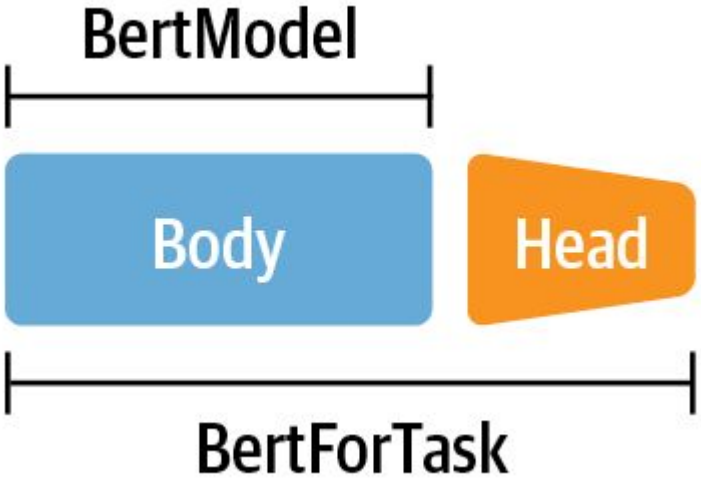
Next Sentence Prediction (NSP)



Fonte: Godoy (2019) BERT | dl-visuals

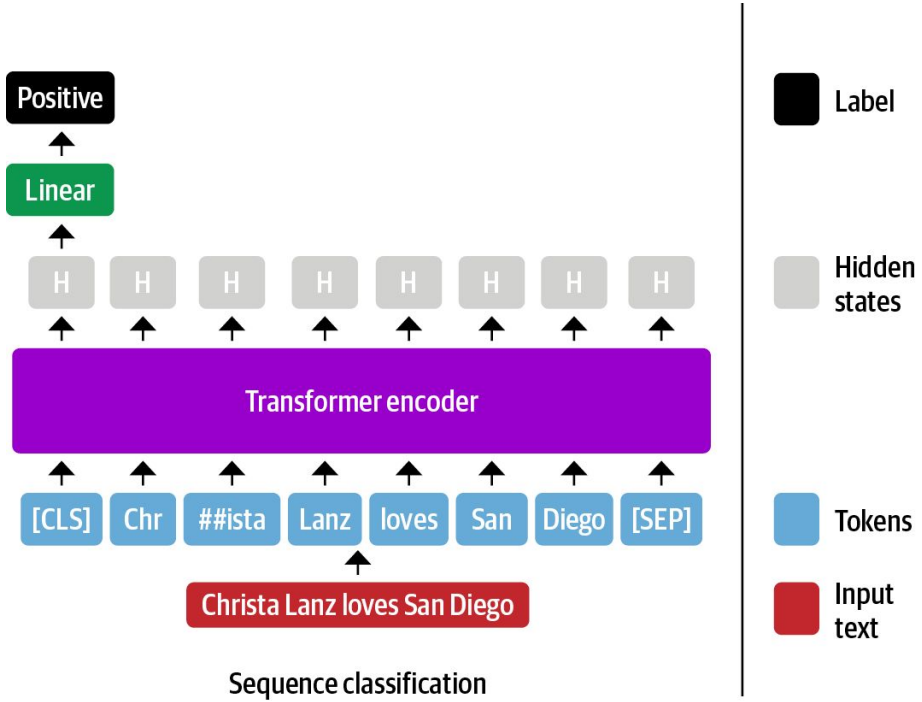
BERT é um modelo flexível para **adaptações**

Adaptação do BERT



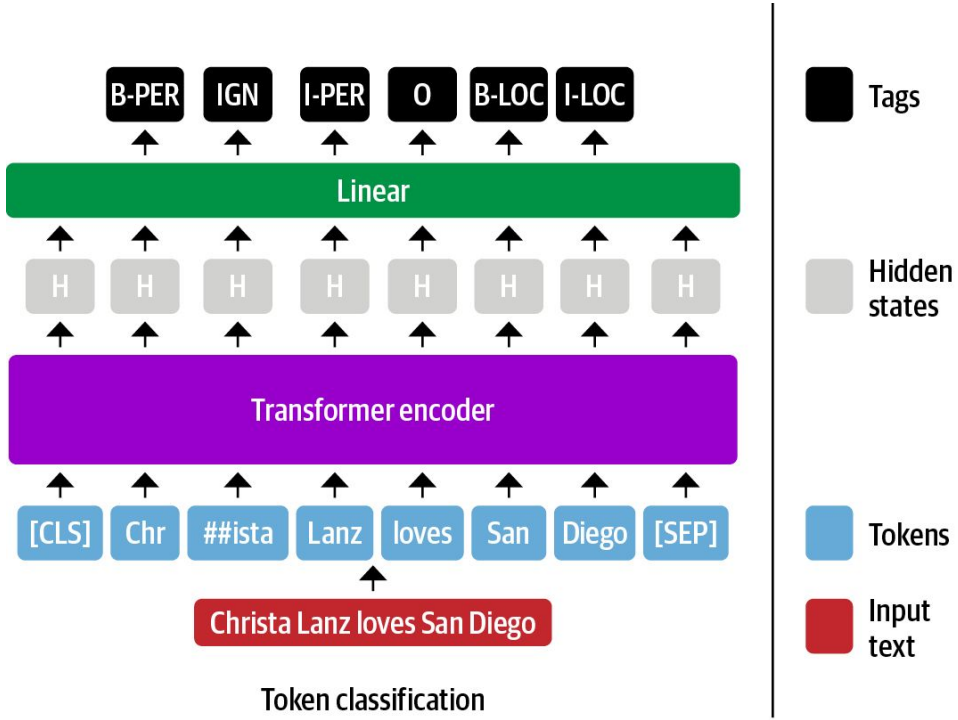
Fonte: Tunstall et al. (2022)

Adaptação para classificação de texto

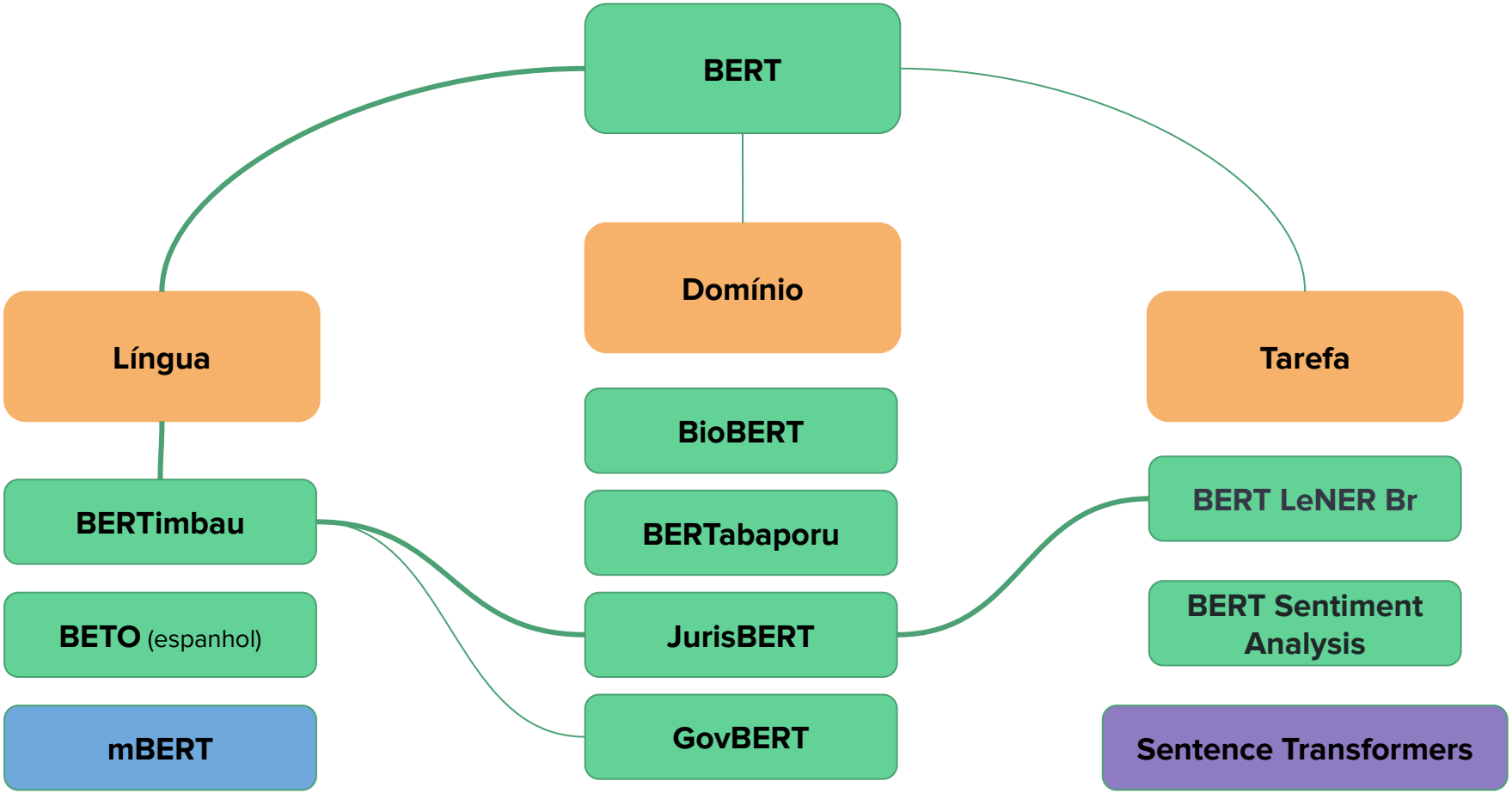


Fonte: Tunstall et al. (2022)

Adaptação para classificação de tokens



Fonte: Tunstall et al. (2022)


BERT
Língua
Domínio
Tarefa
BioBERT
BERTimbau
BERTabaporu
BERT LeNER Br
BERT Sentiment Analysis
BETO (espanhol)
JurisBERT
mBERT
GovBERT
Sentence Transformers

BERTimbau

- Apresentado no BRACIS por **Souza et al. (2020)**.
- Treinado com corpus Portuguese Web Corpus (brWac) + Wikipedia PT, com cerca de 2.7 bi tokens ~= 50Gb.
- Tokenizador WordPiece especificamente treinado com textos em Língua Portuguesa.
- Mantém o treinamento MLM + NSP
- Avaliado especificamente em tarefas para o Português.

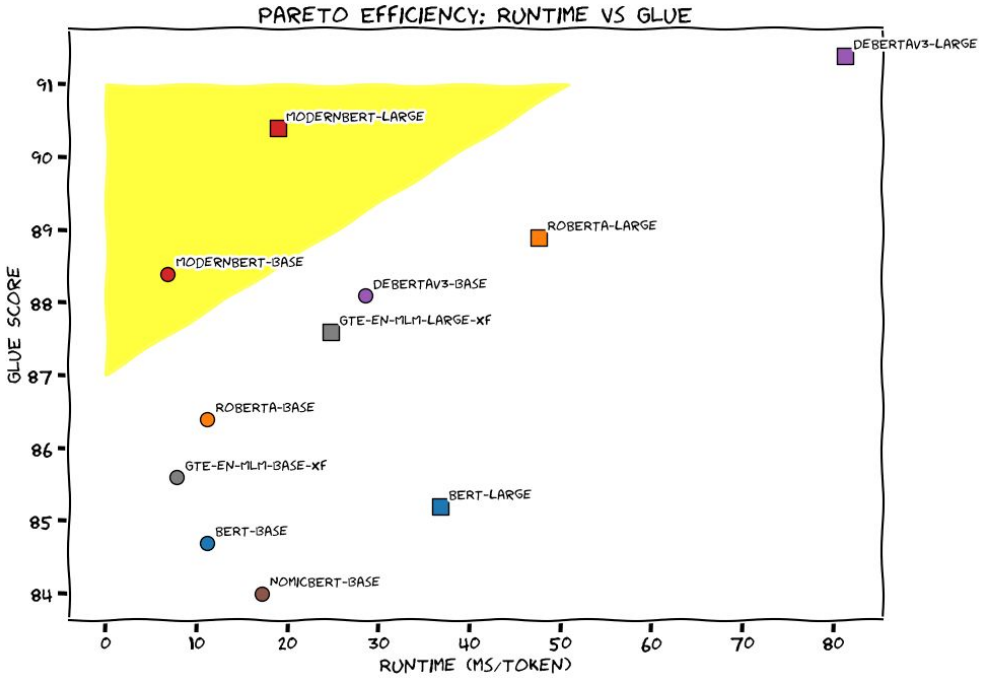


RoBERTa - Robustly Optimized BERT Pretraining Approach (Liu, et al. 2019)

1. Remoção da tarefa de Next Sentence Prediction (NSP)
2. Maior volume de dados para pré-treinamento (160 Gb)
3. Máscara dinâmica dos tokens no treinamento com MLM
4. Batch size e sequências maiores durante o treinamento (até 8.000 exemplos)
5. Maior exploração do espaço de hiperparâmetros
6. Mesma arquitetura

ModernBERT (Warner et al. 2024)

- Tentativa de trazer melhorias das Arquiteturas Decoders para a Arquitetura Encoder
- Rotary Positional Embeddings (ROPE)
- GeGLU funções de ativação
- Flash Attentions Layers
- Sequence Packing
- Treinado com conjunto de dados diversos



Fonte: Warner et al. (2024) Huggingface's blog post announcement.

Demonstração

Obrigado!



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